

The Frozen Transfer Triangle

SAT, Quantum Pointer States, and Fluid Coherence as an Internal Structural-Mode Transfer Test

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2026

Abstract

Papers 52 and 53 tested whether frozen structural weights learned in one internal MAAT benchmark transfer to another without target-domain retuning. Those experiments suggested that the connectivity support V may be a particularly portable low-frequency structural mode, but the evidence was limited mostly to the SAT–quantum axis.

This paper extends the frozen-transfer protocol to a three-domain triangle: SAT frustration fields, quantum pointer-state robustness, and fluid coherence/blow-up diagnostics. Each domain is represented by the same support coordinates $(H, B, S, V, R_{\text{rob}})$, and each target is oriented so that larger quality means greater structural viability. For each source domain we learn non-negative weights once, freeze them, and evaluate all six cross-domain directions without sign flips or target-domain tuning. Equal weights, source-trained NNLS weights, positive ridge weights, V -only target alignment, and shuffled-defect nulls are compared.

The result is informative precisely because it is not a uniform success. Source-trained NNLS transfer remains asymmetric: SAT transfers to quantum ($\rho_S = 0.6034$) and fluids ($\rho_S = 0.6571$), while quantum-to-SAT and fluid-to-SAT fail under NNLS. The V -only score is positive in all three target domains and exceeds the shuffled-defect null, but it must be interpreted as target-mode alignment rather than source-trained transfer. Its six-direction mean $\rho_S = 0.4735$ is row-duplicated by construction and is therefore not a fair architecture-level comparison against NNLS or ridge-positive transfer.

The conclusion is narrow: Paper 66 supports a weak but null-stable connectivity-sensitive mode across these internal domains, but the signal is strong mainly in the support-related Quantum and Fluid targets and weak in the cleaner SAT target. It does not support universal transfer of full source-trained structural architectures. Universality, if present, is more plausibly located in separable structural modes than in one compressed global score.

1 Purpose

Earlier MAAT transfer papers moved the framework from scalar-score comparison toward frozen structural-mode testing. Paper 52 asked whether a defect architecture learned on SAT could transfer to quantum pointer-state robustness without target-domain retuning. Paper 53 showed that the result was robust to several fitting rules but also made clear that transfer is conditional, asymmetric, and sensitive to the chosen sector.

The present paper asks a sharper question:

Is the connectivity mode V the shared transfer channel across SAT, quantum, and fluid benchmarks?

This is not an external validation test. All three domains are repository-internal benchmarks from the existing MAAT series. The purpose is methodological: to determine whether the previously observed V -dominance survives the addition of a third, dynamically very different domain.

2 Frozen Transfer Protocol

Each domain D contains instances x_i , support coordinates

$$(H_i, B_i, S_i, V_i, R_{\text{rob}i}) \in (0, 1]^5,$$

and a predeclared quality target q_i oriented so that larger q_i means greater structural viability. Here H, B, S, V are the primary diagnostic supports. R_{rob} is included only as a derived closure diagnostic coordinate, not as a fifth primitive support sector. The supports are converted to defect coordinates

$$d_a(x_i) = -\log(\epsilon + \Gamma_a(x_i)), \quad a \in \{H, B, S, V, R_{\text{rob}}\}. \quad (1)$$

For a source domain A , weights $\lambda_a^{(A)} \geq 0$ are learned using only the source-domain defects and source-domain quality target. The weights are normalized to $\sum_a \lambda_a = 1$. The frozen score on a target domain B is then

$$S_{A \rightarrow B}(x) = -\sum_a \lambda_a^{(A)} d_a^{(B)}(x). \quad (2)$$

The evaluation statistic is the Spearman rank correlation between $S_{A \rightarrow B}$ and the target quality $q^{(B)}$. No sign flips, target-domain regression, or target-domain reweighting are allowed.

2.1 Fit Rules

Four frozen rules are tested:

- **equal:** equal weighting over the five diagnostic coordinates $(H, B, S, V, R_{\text{rob}})$;
- **NNLS:** non-negative least squares fit to source defect cost;
- **ridge-positive:** ridge-regularized source fit clipped to non-negative weights;
- **V-only:** $\lambda_V = 1$, all other weights zero.

The V -only rule is not a learned transfer architecture. It is a direct diagnostic of whether the connectivity coordinate itself is aligned with structural quality in each target domain. This distinction is essential: a strong V -only result supports a shared mode, not necessarily full source-specific transfer.

3 Domains

Domain	Instances	Quality target
SAT	550	1 – minmax(log DPLL nodes)
Quantum	3500	pointer-state robustness target
Fluid	340	1 – minmax(MAAT warning)

Table 1: The three internal domains used in the frozen-transfer triangle.

The SAT domain is inherited from the Paper-50b frustration-field benchmark. The quantum domain is inherited from the Paper-51 non-commuting Lindblad pointer-state benchmark. The fluid domain combines the Paper-60 Burgers shock-warning and 2D Navier–Stokes regular-control diagnostics.

Fluid-target caveat. The Fluid quality target is not independent of the MAAT support language. It is constructed from the Paper-60 warning diagnostic, which itself depends on support-like quantities including the connectivity channel V . Transfers into Fluid are therefore partly expected to favour support-aligned or V -weighted scores. For this reason, the cleanest stress direction in this paper is transfer *into* SAT, where the target is DPLL hardness rather than a support-derived fluid warning.

Primary stress-test convention. In the interpretation below, transfer into SAT is treated as the primary stress test because its target variable is least derived from the MAAT support language. Quantum and Fluid remain useful internal targets, but they are closer to the support construction and therefore carry a higher risk of internal alignment.

4 Frozen Weights

Source	Rule	λ_H	λ_B	λ_S	λ_V	λ_R
SAT	NNLS	0.000	0.000	0.000	1.000	0.000
Quantum	NNLS	0.484	0.000	0.298	0.218	0.000
Fluid	NNLS	0.000	0.105	0.475	0.420	0.000

Table 2: Source-trained NNLS weights. SAT collapses to the V -mode, while Quantum and Fluid use mixed modes.

The SAT result reproduces the earlier Paper-52/53 observation: the SAT source fit is connectivity-dominated. The added Fluid domain is not purely V -dominated; it mixes activity and connectivity.

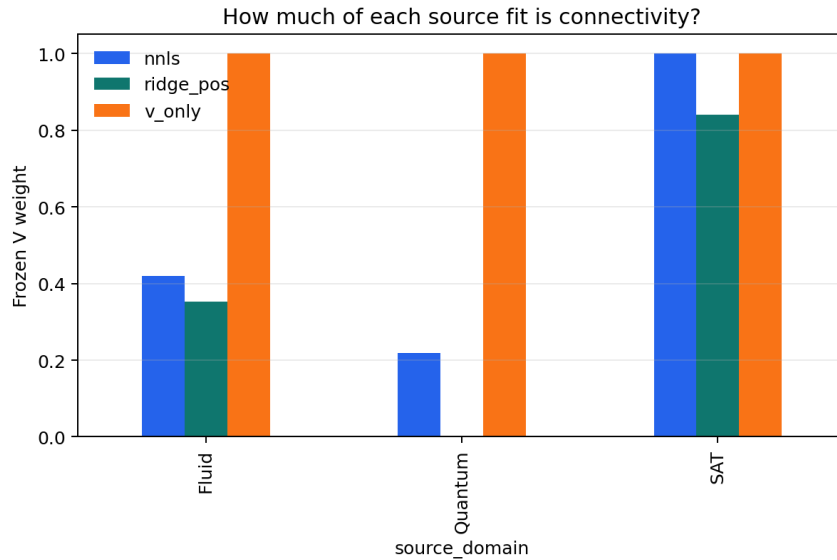


Figure 1: Frozen V -weight by source and fit rule. The SAT source is almost purely connectivity-driven, while the Fluid and Quantum source fits distribute weight across additional modes.

5 Results

5.1 NNLS Transfer Triangle

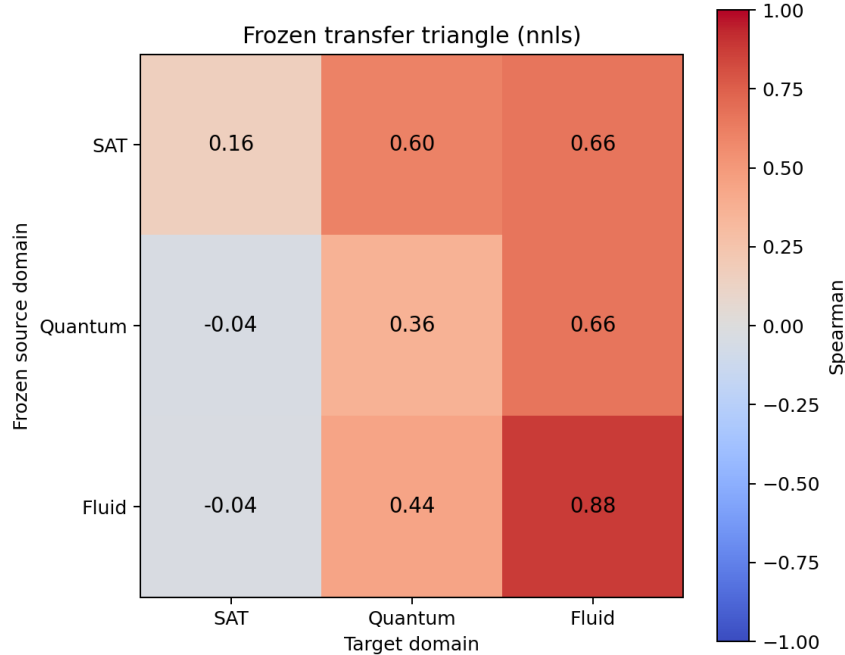


Figure 2: Frozen NNLS transfer triangle. Rows are source domains, columns are target domains. Transfer into Fluid is strong but partly expected because the Fluid target is support-derived. The cleaner stress test is transfer into SAT: Quantum and Fluid source fits do not transfer there.

Transfer	ρ_S	Null q_{95}	Above null?
SAT $\rightarrow$ Quantum	0.6034	0.0287	yes
SAT $\rightarrow$ Fluid	0.6571	0.0923	yes
Quantum $\rightarrow$ SAT	-0.0408	0.0687	no
Quantum $\rightarrow$ Fluid	0.6635	0.0847	yes
Fluid $\rightarrow$ SAT	-0.0360	0.0622	no
Fluid $\rightarrow$ Quantum	0.4413	0.0270	yes

Table 3: Six cross-domain directions under source-trained NNLS weights. The SAT target remains the hardest to reach from non-SAT source architectures.

NNLS does not produce universal source-trained transfer. It succeeds in four of six directions but fails when Quantum or Fluid weights are transferred into SAT. This is consistent with earlier SAT results: SAT hardness depends on local frustration geometry, and non-SAT global modes do not automatically capture that geometry. The SAT $\rightarrow$ Fluid NNLS result is also not independent of V -only target alignment, because the frozen SAT NNLS weights collapse to $\lambda_V = 1$ in Table 2. The Fluid $\rightarrow$ Quantum result is positive but modest ($\rho_S = 0.4413$), and should also be read cautiously: the Fluid source weights are learned from a support-derived warning target, so this direction does not provide a clean independent test of structural transfer.

5.2 V-Only Target Alignment

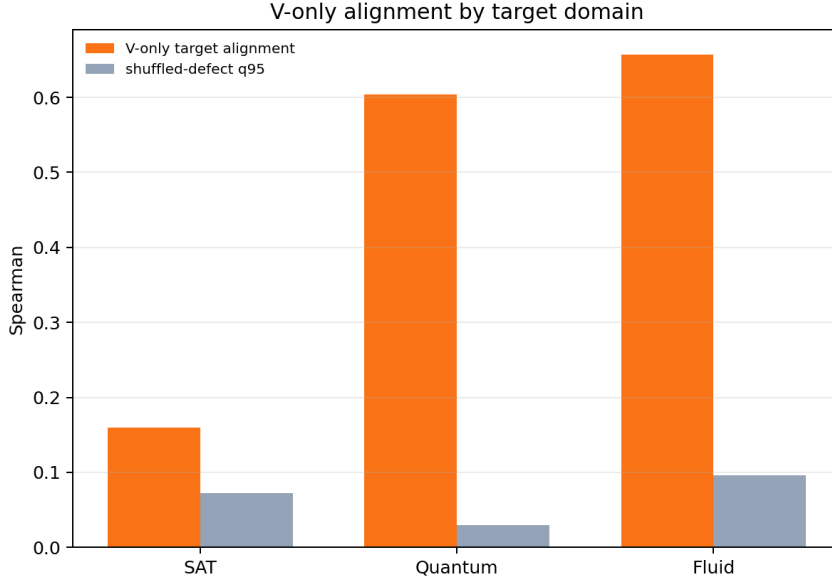


Figure 3: V -only target alignment. This is not a transfer triangle: the same V -only score is evaluated against each target-domain quality variable. Bars compare observed alignment with the conservative shuffled-null q_{95} threshold.

Target	$\rho_S(V, q)$	Null q_{95}	Above null?
SAT	0.1600	~ 0.072	yes
Quantum	0.6034	~ 0.030	yes
Fluid	0.6571	~ 0.096	yes

Table 4: Target-domain V -mode alignment. The SAT signal is weaker but still above the shuffled-defect null. Quantum and Fluid show strong connectivity alignment.

This is the central mode-level result of Paper 66. The V -only score is positive and null-stable in all three target domains. However, the six-direction V -only mean repeats the same target alignment twice per target because V -only has no source dependence. It is therefore a target-mode stability statistic, not a fair competitor to six source-specific NNLS or ridge-positive transfers.

Rule	$\langle \rho_S \rangle_{\text{cross}}$	Interpretation
equal	0.3666	equal diagnostic-coordinate baseline
NNLS	0.3814	source-trained non-negative architecture
ridge-positive	0.3979	regularized source-trained architecture

Table 5: Mean cross-domain summary for architecture-level rules. The V -only diagnostic is intentionally excluded because it is a row-duplicated target-alignment statistic rather than a source-trained architecture.

For audit transparency, the row-duplicated V -only six-direction mean is $\langle \rho_S \rangle_{\text{cross}} = 0.4735$. It is not included in Table 5; the meaningful V -only information is the target-domain alignment reported in Table 4.

The ridge-positive rule slightly outperforms NNLS (0.3979 versus 0.3814). This is a small but useful diagnostic result: regularization appears to damp source-domain overfitting and improves

transfer relative to the unregularized NNLS source architecture. That behaviour is consistent with the central interpretation of this paper: portable low-frequency modes are more stable than fully source-adapted architectures.

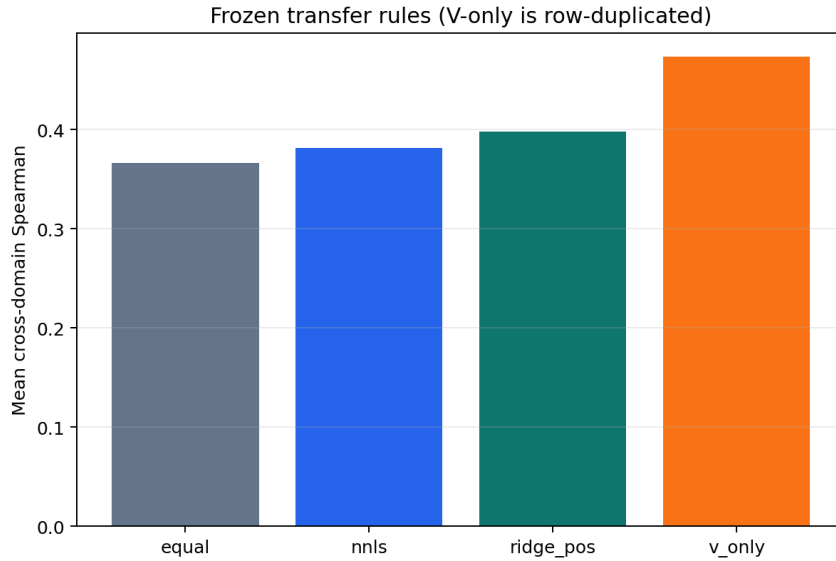


Figure 4: Mean cross-domain Spearman score across six directions. The V -only bar is a row-duplicated target-alignment statistic, not a source-trained transfer rule. The figure is retained to show why the connectivity coordinate should be separated from full-architecture transfer.

5.3 Shuffled-Defect Nulls

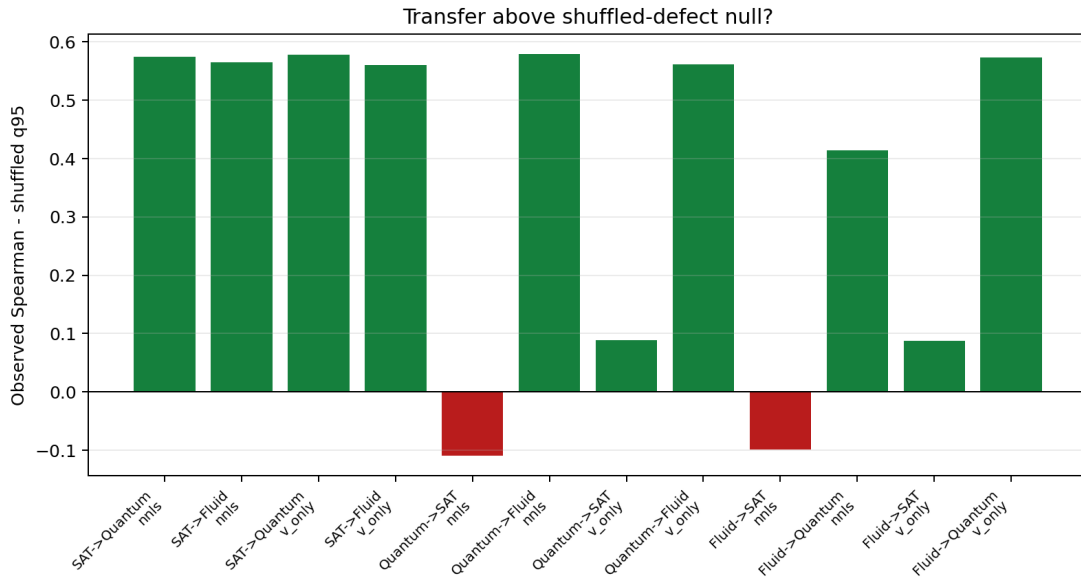


Figure 5: Observed transfer score minus shuffled-defect q_{95} . Green bars exceed the null threshold. NNLS fails for Quantum $\rightarrow$ SAT and Fluid $\rightarrow$ SAT. The V -only entries remain above the shuffled null but are duplicated target-alignment tests rather than source-trained transfers.

The shuffled-defect nulls preserve the marginal distributions of the support sectors but destroy row-level sector coherence. Exceeding this null therefore means that the score uses coherent

support structure rather than merely exploiting a one-dimensional marginal.

6 Interpretation

Paper 66 supports a precise, limited conclusion:

The connectivity coordinate V behaves like a weak shared low-frequency mode across SAT, quantum pointer states, and fluid coherence diagnostics, with strong alignment in the support-related targets and only weak alignment in SAT.

This does not mean that MAAT has demonstrated universal cross-domain transfer. The source-trained NNLS architectures remain asymmetric and fail in important directions. In particular, non-SAT source weights do not transfer to SAT under NNLS. This reinforces the lesson from Papers 50b, 55, and 57: SAT hardness contains local frustration geometry that is not captured by generic global support mixtures.

The strongest positive interpretation is therefore not:

“one MAAT score transfers everywhere.”

It is:

“some structural modes transfer more reliably than full architectures.”

This is consistent with the v1.4/v1.4.1 shift from global scalar compression to multi-scale, partially separable defect modes. The V -mode appears to be one such low-frequency mode, but the SAT result ($\rho_S = 0.1600$) shows that this mode is weak in the least support-derived target. The conservative interpretation is therefore not that V transfers strongly everywhere, but that it remains detectable under shuffled-defect nulls while full source-trained architectures fail in the cleanest SAT stress direction.

7 Limitations

- All domains are internal MAAT benchmarks; this is not external validation.
- The Fluid target uses the Paper-60 MAAT warning diagnostic, so the Fluid-domain quality variable is not independent of the support language. Transfers into Fluid should therefore be treated as partly circular and less diagnostic than transfer into SAT.
- The V -only rule is not source-trained transfer. It is a target-mode alignment test, and its six-direction mean is row-duplicated by construction.
- Equal weights are taken over five diagnostic coordinates; R_{rob} remains a derived closure diagnostic rather than a primitive support sector.
- SAT uses DPLL-node hardness inherited from earlier benchmarks rather than modern CDCL industrial-instance hardness.
- The benchmark tests three domains only. A true universality claim would require external domains and predeclared transfer rules.

8 Reproducibility

Repository:

<https://github.com/Chris4081/structural-selection-principle>

Run:

```
cd experiments/frozen_transfer_triangle_paper66
python3 frozen_transfer_triangle_paper66.py
```

Generated outputs:

- `paper66_transfer_results.csv`,
- `paper66_frozen_weights.csv`,
- `paper66_shuffle_nulls.csv`,
- `paper66_summary.json`,
- five PNG figures, including the V -only target-alignment plot.

Data and attribution note. Paper 66 reuses repository-internal derived outputs from Papers 50b, 51, and 60. No new external dataset is introduced in this benchmark. The derived CSV/JSON/PNG outputs are reproducibility artifacts and should be distinguished from the original scripts and prior paper outputs.

9 Conclusion

Paper 66 converts the earlier two-domain transfer result into a three-domain triangle. The result is mixed but informative. Full source-trained architectures do not transfer universally. However, when treated as a target-alignment diagnostic rather than a source-specific transfer rule, the V -connectivity coordinate is positive and null-stable across SAT, quantum, and fluid target domains. The important qualification is that the SAT alignment is weak ($\rho_S = 0.1600$), while the stronger Quantum and Fluid alignments occur in targets closer to the MAAT support language.

The useful lesson is therefore:

universality, if present, appears first as conditional and domain-dependent mode portability, not as a compressed global MAAT score.

References

- [1] C. Krieg, “SAT Frustration-Field Compression Test,” preprint (2026).
- [2] C. Krieg, “Quantum Pointer-State Selection,” preprint (2026).
- [3] C. Krieg, “Cross-Domain Transfer of Structural Selection,” preprint (2026).
- [4] C. Krieg, “Structural Mode Transfer Robustness,” preprint (2026).
- [5] C. Krieg, “Fluid Coherence and Blow-up Diagnostics,” preprint (2026).